Fiscal Year:	FY 2006	**Task Last Updated:**	FY 12/29/2006
PI Name:	Gould, Harvey Allen Ph.D.		
Project Title:	Feasibility Study for an Electron Electric Dipole Moment		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Laser cooled atomic physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	gould@lbl.gov	**Fax:**	FY 510-486-7981
PI Organization Type:	GOVERNMENT	**Phone:**	510-486-7777
Organization Name:	Lawrence Berkeley National Laboratory		
PI Address 1:	MS 71-259		
PI Address 2:	LBNL		
PI Web Page:			
City:	Berkeley	**State:**	CA
Zip Code:	94720	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	12/01/2003	**End Date:**	09/30/2007
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
Contact Monitor:	Maleki, Lute	**Contact Phone:**	818-354-3688
Contact Email:	lute.maleki@jpl.nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 9/30/07, per PI and GRC (7/07)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	C73534 A		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	To determine if a slow, laser-cooled atom microgravity experiment can improve the sensitivity to an electron electric dipole moment (EDM) by orders of magnitude. Develop and test experimental features such as electric field quantization, multiple quantum transitions, beam focusing and reflection, electric field ramping, pulsed oscillatory fields, state selection and Stark separated levels. Detemine sensitivity to systematic effects, sensitivity to electric and magnetic noise. Will determine the most useful features to incorporate, identify problem areas, and help assess the feasibility of proceeding to flight definition studies for a microgravity electron electric dipole moment experiment.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Spinoff can improve accuracy of cesium atomic clocks
Task Progress:	The data from a demonstration Cs fountain electron EDM experiment has been collected and analyzed and a paper is being prepared for submission to Physical Review A in 2007.
Bibliography Type:	Description: (Last Updated: 02/25/2014)
Abstracts for Journals and Proceedings	Gould H, Amini JA, Munger C. "Prospects for Testing Standard Model Extensions with Electron Electric Dipole Moment (EDM) Experiments." 2006 April Meeting of the American Physical Society, Dallas TX April 22-25, 2006. 2006 April Meeting of the American Physical Society, BAPS abstract I9.00008, 2006. , Apr-2006
Abstracts for Journals and Proceedings	Gould H, Amini JA, Munger CT. "Demonstration Cold Atom Fountain Electron Electric Dipole Moment (EDM) Experiment." April meeting of the American Physical Society, Dallas TX, April 22-25, 2006. April meeting of the American Physical Society, BAPS abstract D1.00078, 2006. , Apr-2006
Abstracts for Journals and Proceedings	Gould H, Amini JA, Munger CT. "Demonstration of an Electron Electric Dipole Moment Experiment Using Electric-Field Quantization in a Cesium Cold Atom Fountain." Annual Meeting of the Division of Atomic Molecular and Optical Physics, Knoxville TN, May 16-20, 2006. Annual Meeting of the Division of Atomic Molecular and Optical Physics, BAPS abstract W1.00053, 2006. , May-2006
Articles in Peer-reviewed Journals	Amini JM, Munger CT Jr, Gould H. "Electron electric-dipole-moment experiment using electric-field quantized slow cesium atoms." Physical Review A, submitted December 2006. , Dec-2006